4 *Refining the best result with improvement-based policies*

This chapter covers

- The Bayesian optimization loop
- The tradeoff between exploitation and exploration in a Bayesian optimization policy
- Improvement as a criterion for finding new data points
- Bayesian optimization policies that leverage improvement

In this chapter, we first remind ourselves of the iterative nature of BayesOpt: we alternate between training a Gaussian process on the collected data and finding the next data point to label using a BayesOpt policy. This forms a virtuous cycle where our past data inform future decisions. We then talk about what we look for in a BayesOpt policy, a decision-making algorithm that decides which data point to label. A BayesOpt policy needs to balance between sufficiently exploring the search space and zeroing in on the high-performing regions.

Finally, we learn about two policies that seek to improve from the best-seen data point so far in the BayesOpt loop: *Probability of Improvement* and one of the most commonly used BayesOpt policies, *Expected Improvement*. For example, in a hyperparameter tuning application where we want to identify the neural network that gives the highest validation accuracy on a data set, and the highest accuracy we have seen so far is at 90 percent, the policies we will learn in this chapter try to *improve* from this 90 percent threshold. In the material discovery task we saw in Table 1.2, where we'd like to search for metal alloys that mix at low temperature (corresponding to high stability), and 187.24 is the lowest temperature we have found, the two aforementioned policies will seek to find values lower than this 187.24 benchmark.

Amazingly, thanks to the Gaussianity of our belief about the objective function, how likely or how much we can expect to improve from the best-seen point may be computed in closed form. This is to say that while we don't know what the objective function looks like at unseen

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>